AI for SMEs — Handbook

This edition covers the challenges and opportunities of implementing AI solutions in regional manufacturing and wholesale SMEs, focusing on specific issues highlighted by [Company_1] B.V. in their third quarter report.

The questions this handbook answers

#	Theme	The SME-facing question
1	Operational Gaps & Data Realities	How can we reduce implementation times for new clients and manage energy costs at data centers?
2	Practical Implementation & ROI	What AI tools and strategies should [Company_1] B.V. use to improve customer retention, operational efficiency, and cost control?

Chapter template: Theme 1 — Operational Gaps & Data Realities

1. The question, for this kind of SME

How can we reduce the average implementation time from 48 days to the target of 35 days for new clients?

2. The translation

The company is facing significant delays in implementing their automated logistics platform due to supply chain disruptions and hardware supplier issues. This has led to an increase in client acquisition times, which impacts customer satisfaction and revenue growth.

3. Illustrative use case (CRISP-DM)

Business understanding

The goal is to reduce the average implementation time for new clients from 48 days to 35 days by identifying bottlenecks in the supply chain process and streamlining operations.

Data understanding

Data sources include order logs, ERP tracking systems, and cloud infrastructure records. The specific data needed are transaction records, order logs, and ERP system integration details.

Data preparation

The data will be cleaned to remove duplicates, inconsistencies, and missing values. Anonymization will ensure compliance with GDPR regulations while preserving the necessary information for analysis.

Modelling

A time-series forecasting model can predict future implementation times based on historical trends. Regression models can also identify correlations between specific variables (e.g., supplier delays, cloud infrastructure performance) and project timelines.

Evaluation

Success will be measured by reducing the average implementation time to 35 days or less within a six-month period. Key metrics include cost savings from faster implementations and improved customer satisfaction due to quicker service delivery.

Deployment

The automated recommendations will be integrated into existing ERP systems, allowing warehouse managers to receive real-time alerts on potential delays and take corrective actions proactively.

4. Recommended tooling

- **Power Query**: For data cleaning and integration (estimated effort: 10 hours, cost: €500)
- **Python Script**: To automate the forecasting model (estimated effort: 20 hours, cost: €800)

5. Responsible AI & trust

Data security will be ensured through anonymization techniques and GDPR compliance measures. Local data centers will be used to minimize energy costs while maintaining performance standards.

6. First five steps for an owner

1. Conduct a thorough analysis of the current supply chain process.

2. Identify key suppliers and their impact on implementation times.
3. Implement stricter cost control measures in cloud infrastructure and server usage.
4. Develop a strategic partnership plan with new clients to reduce delays.
5. Train staff on AI tools and best practices for efficient use.

7. Sources

- [Company_1] B.V. Third Quarter Report, October 2026
- Industry benchmarks on supply chain disruptions and cost control measures

Tooling index

Tool	Use Case	Estimated Effort (hours)	Cost Estimate (€)
Power Query	Data Cleaning & Integration	10	500
Python Script	Forecasting Model Development	20	800

Regional ecosystem & support

- Oost NL: Offers resources and training for SMEs in logistics and supply chain management.
- VNO-NCW: Provides financial assistance and guidance on cost control measures.
- HAN Applied Research: Offers consulting services and research projects tailored to regional SME needs.

About this handbook

This handbook is written by students of HAN University of Applied Sciences, teams collaborating across the AIBS (Applied AI Business Solutions) and AEL (Applied Engineering Learning) modules. The quality bar is set high for clarity, accuracy against appraised sources, responsible-AI coverage, and usefulness. This handbook serves as educational inspiration rather than formal business liability. Real SME cases are anonymised or published with consent.